

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-aba44318d3410bdc9.jsonl`  
- SHA-256: `43cd695d78b5e8316f97ef45130d7ca05e1347130ee23bd124edde1247647a90`  
- Source modified: `2026-06-10T05:44:09+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_2a64bc88-2a8`  
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`
```

Transcript

1. user (2026-06-10T05:42:43.052Z)

CONTEXT ? two sibling repos on a Windows machine:

- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).

- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).

Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources. ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.

QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.

ROLE: adversarial verifier. An auditor ("docs-strategy") claimed the finding below. Try to REFUTE it against the actual files. Default to isReal=false if the evidence does not hold up, the behavior is documented/deliberate (e.g. listed in docs/CODE_MAP.md gotchas or DEFERRED_HARDENING_PLAN.md as a known deferral ? unless the auditor adds genuinely NEW consequence), or the severity is materially overstated.

FINDING: {"title":"WASB C3 'train-own-weights' mitigation cites a ROADMAP plan that does not exist (and cannot, per ADR-002)","severity":"high","area":"WASB checkpoint provenance (C3)","file":"C:/Users/avidu/Projects/badminton-highlight-indexer/docs/COMMERCIALIZATION.md:42","evidence":"COMMERCIALIZATION.md C3 (line 42) gives the mitigation 'Train-own weights (**Phase-4 roadmap**) or confirm per-checkpoint terms'; OWNED_MODEL_TRAINING_STUDY.md:125-126 says 'ROADMAP Phase-4 \"train own WASB weights\" is a **prerequisite, not an aside**'. But ROADMAP.md contains zero mentions of training weights (grep 'weights|train our own' = no matches) ? its Phase 4 is a learned TEMPORAL segmenter built ON TOP of WASB output. ADR-002 (docs/archives/decisions/DECISIONS.md:69-71) explicitly states the collector annotations 'are temporal-only... they **cannot train the shuttle detector** (TrackNetV4/WASB...), which needs coordinate labels.' Meanwhile ADR-005 (DECISIONS.md:244) headlines 'WASB-SBDT and TrackNetV3 are MIT (CLEAR for commercial use) ? this resolves the licensing unknown', with the checkpoint caveat only in a trailing clause; CODE_MAP.md:110 describes wasb_hybrid as '(MIT; ships pretrained badminton weights, the LIVE default)' with no provenance caveat. The plan (M0.3 ii, lines 118-120) correctly makes C3 a ship-gating blocker ('multiplies per sport', owner: yes), but no document anywhere specifies HOW own weights would be trained (per-frame x/y labels do not exist in the corpus).","why_it_matters":"The documented escape hatch for a ship-gating commercial blocker points at a phase that cannot deliver it. A reader could believe own-weights training is already planned and deprioritize C3; in reality resolving it requires a new, un-scoped workstream (coordinate labeling or a clean detector swap)

that grows with every sport WASB is reused for ? including the new table-tennis use."}

Steps: open the cited file(s) and any counter-evidence files; verify quotes/line claims; judge severity honestly. Cheap greps allowed; do NOT run the pipeline. Return isReal, confidence, and a 1-3 sentence verdict (with corrected detail if partially right).

2. user (2026-06-10T05:42:53.315Z)

Priority: **P0** (do first) . **P1** . **P2** (lowest). Rows are grouped **Pending ? In-flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply order.)

```
| # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
|---|---|---|---|---|---|
| | **? PENDING (needs action / not started)** | | | |
| C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 + AAC**
('libx264' in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for
HW encode. See §4. Decision: provision Ada-class GPU? |
| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `api/process` blocks on
long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md`
(pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md`
(#16). |
| | **? IN-FLIGHT (in progress / watch)** | | | |
| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini 2.5
Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open
VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A. |
| C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF_MARKET_RESEARCH.md`)
recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests:
10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur
clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See §6. |
| C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected for
an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm
quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
| C3 | **Shuttle-tracking weights provenance** | **P1** | ? ?? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or**
confirm per-checkpoint terms ?? before commercial deploy. |
| C14 | **Owned-model base licensing + training-data policy** | **P1** | ? ?? | Owned video-QA
model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ?
**Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep
weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent
Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train
on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the
**consented dataset + eval harness**, not the model. | Lock to Apache-2.0; **open-model teachers
only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-
clip consent ledger. See PLATFORM §5A/§7. |
| C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads is
unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm
decode-side exposure with counsel before launch. |
| C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~$0.05
input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs
unmeasured. | **Benchmark $/video & latency on the target VM** for A/B/C before pricing. See §5.
|
| C10 | **Value layer (product surface)** | **P2** | ? | Minimal exports shipped (EDL/JSON/CSV +
`basic_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats,
clip gallery. See §6. |
| C11 | **Input-quality enforcement** | **P2** | ? | Per-video observability reports surface a
customer verdict + footgun flags incl. audio-present (PR #50). | Turn more
`VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules. |
| C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned `<0.30.0` (deprecation ahead). |
```

```
Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |
| | **? COMPLETED** | | | | |
| C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralalytics` YOLOv8 replaced
with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL code/weights in
the stack. | Done. Keep new deps permissive. |

---
```

2. Licensing & dependency posture (refreshed risk table)

Verdicts updated against the current stack (`requirements.txt` + code). Original 2026-05-30 verdicts and the 25-claim verification transcript are in `archives/MONETIZATION\_AUDIT.md`.

Dependency	License	SaaS obligation?	Current verdict	Notes
------------	---------	------------------	-----------------	-------

3. user (2026-06-10T05:42:54.182Z)

```
Exit code 2
sed: can't read C:/Users/avidu/Projects/badminton-highlight-
indexer/docs/OWNED_MODEL_TRAINING_STUDY.md: No such file or directory
```

4. user (2026-06-10T05:43:01.212Z)

No files found

5. user (2026-06-10T05:43:02.455Z)

No matches found

6. user (2026-06-10T05:43:08.461Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
```

7. user (2026-06-10T05:43:09.170Z)

```
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:126:  weights" is a **prerequisite,
not an aside.**
```

8. user (2026-06-10T05:43:16.173Z)

- **Exclude minors by default + separate training opt-in;** per-user adapters trained only on that user's consented video (deletion = drop the adapter). Enforced at the corpus-membership layer before transcoding.
- **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any cloud *training* needs a signed DPA + no-train assurance; route only owned/consented footage, never user uploads.

Cost (spend on labels, not GPUs)

- **Gen-0:** ~\$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).
- **Gen-2 LoRA:** ~\$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token cost on real 1080p/60fps clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together) ~\$2?30 *if* text-priced (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).
- **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini (~\$0.05/10-min clip) **only** above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now, owned-

primary later.

****Benchmark \$/call before committing** (COMMERCIALIZATION C7).**

Open risks & critic caveats (read before acting)

- ****?? WASB checkpoint provenance (C3) propagates.**** The study calls WASB "MIT," but that's the ***code***; the distributed ``wasb_badminton_best.pth.tar`` is ****academic-data-trained, provenance unresolved**** (COMMERCIALIZATION C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label teacher ? a "clean" owned model trained on provenance-unclear features ****does not launder the provenance.**** ROADMAP Phase-4 "train own WASB weights" is a ****prerequisite, not an aside.****
- ****Gemma 4 "clean + native-video" rests on post-cutoff secondary sources**** ? keep AMBER until a raw HF LICENSE revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is less documented than Qwen's.
- ****Existing distilled model was trained on Gemini silver labels**** ? the (correct) eval-only policy now ****forbids** this; that pipeline must be purged of Gemini-derived training labels** to comply.
- ****Policy ratifications needed:**** guardrail literally says "Apache-2.0 only" but the approved list includes MIT ? ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at our scale.
- ****On-device privacy claim is unbenchmarked**** ? no tokens/sec for a quantized 4B VLM on 8 GB Turing at 1080p/60fps. ****Measure before relying on it.****
- ****Amplification is load-bearing for Gen-2**** ? 128 raw rallies can't reach the ~5k?20k clip?QA band alone; if augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed. Amateur 60fps may need more labels than broadcast-trained literature implies.
- ****Dropped option worth keeping:**** InternVideo2 encoder (MIT) + light grounding head ? the ***only* data-realistic** owned-video experiment available at n?128 today. And: a ****one-time ~\$1?2k 24 GB local card**** (e.g. used 3090) would preserve the privacy lever for ***training*** and dissolve most of the cloud-DPA analysis ? uncoded here.

9. user (2026-06-10T05:43:17.122Z)

```
34:## The four phases
36:### Phase 1 ? Bridge the repos (DONE)
43:### Phase 2 ? Acquire full-resolution video (DONE)
52:### Phase 3 ? Evaluate + tune (IN PROGRESS)
53:`run_phase3_eval.py` walks the manifest, processes each video, and scores detected
61:### Phase 4 ? Learn an offline, API-free segmenter (NEXT)
81:*learned timbre classifier* ? which, like Phase 4, is **gated on golden labels**.
84:The recurring blocker across Phases 3?4 and the audio work is the same: **more
102:| Phase | What | Status |
106:| 3 | Eval + tuning loop (`run_phase3_eval.py`) | ? In progress |
110:> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand regression testing)
```

10. user (2026-06-10T05:43:23.159Z)

F1, mean IoU, boundary error), for both the ``candidates`` (raw detector) and ``final`` (refined) stages. ? ***Why:** the candidates-vs-final split tells us ***where*** error lives ? a ****detection**** problem (candidates miss rallies) vs. a ****segmentation**** problem (candidates catch them, refinement is loose). That diagnosis decides where effort goes, and establishes the baseline any trained model must beat.

Phase 4 ? Learn an offline, API-free segmenter (NEXT)

Replace the paid Gemini refinement with a **lightweight classifier** trained on the rally annotations, consuming features the pipeline already computes and stores (`candidate_segments.stats`). No external API, no new labeling. ? **Why:** this is the payoff ? a free, offline, self-improving rally mode. The **substrate** was chosen from data: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate stage at **recall ~0.05 @ IoU 0.5** on `shuttleset_8` (median best coverage of a rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

Complementary local cues (beyond the segmenter swap)

Once the substrate is WASB shuttle-motion, the next lever is **fusing additional fully-local cues** to raise rally recall/precision without reintroducing a paid API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**, accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in `backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ? verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover WASB-missed rallies) but the classical-onset precision ceiling (~2.2x) is too weak to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a **learned timbre classifier** ? which, like Phase 4, is **gated on golden labels**. See `AUDIO_E1_FINDINGS.md`. Pose/court serve-gating is parked as the #2 cue.

The recurring blocker across Phases 3?4 and the audio work is the same: **more golden labels** is the single highest-leverage move (see the cross-cut note below).

How this shapes the final tool

- **Offline & free by default.** The production rally mode becomes a local model; Gemini is kept only as an upper-bound reference, not a dependency.
- **Self-improving.** Every newly curated, annotated match is more training signal; re-training is a local, deterministic step.
- **Measurable.** Every change is scored against the golden set, so "better" is a number, not a vibe.
- **Sport-generic.** The label ? feature ? classifier machinery and the export bridge already span sports; adding a sport is data, not new architecture.
- **Commercially clean.** Only `is_commercial` data flows through the bridge, so the trained model and any derived product rest on a licensable foundation.

Status at a glance

Phase	What	Status
1	Export bridge (<code>curate export</code> ? manifest + CSVs)	? Done
2	1080p acquisition (<code>fetch</code> , in-place, cookies)	? Done (22/22)
3	Eval + tuning loop (<code>run_phase3_eval.py</code>)	? In progress
4	Offline learned segmenter (WASB shuttle-motion substrate)	? Next (substrate decided: WASB, ADR-004)
+	Audio hit-detection cue (<code>backend/pipeline/audio/</code> , ADR-007)	? E1 probe run ? AMBER, gated on golden labels

> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand regression testing)
> is produced via a vendor process + pluggable annotation provider ? see `[GOLDEN_SET_VENDORING.md]` (`GOLDEN_SET_VENDORING.md`) and **ADR-006**. As of 2026-06-02 the annotation
> guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is selected for an
> initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality requirements
> that bound achievable precision/recall are tracked in `[VIDEO_RECORDING_GUIDELINES.md]` (`VIDEO_RECORDING_GUIDELINES.md`).

ADR-002: Use the sports-data-collector's rally annotations as a golden eval + temporal-segmenter training set

Date: 2026-05-26

Status: Accepted

Context

A sibling project, **sports-data-collector**, curates sports videos and stores per-rally `[start, end]` timestamps (plus stroke counts, scores) in an SQLite DB, with a commercial-license flag per source. We asked whether this data could improve the indexer's rally detection, and for which task.

Key findings:

- The annotations are **temporal-only** (when rallies start/end). They do **not** contain per-frame shuttle `(x, y)` coordinates, so they **cannot** train the shuttle detector (TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
- They are an excellent **golden evaluation set** and a **training signal** for the **temporal rally segmenter** (active vs. dead play over time) ? the stage that is today a hand-tuned heuristic, not a learned model.
- The collector's five sports tables (`badminton_rallies`, `tennis_points`, `cricket_deliveries`, `pickleball_rallies`, `tabletennis_rallies`) all share a `(video_id, ordinal, start_time, end_time)` shape, and the indexer already has `sport_registry` / `segmenter_registry`. So a **generic** `(start, end)` bridge serves every sport.
- The two repos key videos differently: the collector by a human `video_id`, the indexer by **file MD5**. A re-encode/re-download changes the bytes, hence the MD5.

Decision

1. Treat the collector's rally timestamps as (a) the **golden eval set** for rally **segmentation** and (b) the **labels** for training a temporal segmenter ? explicitly **not** as shuttle-detector training data.
2. Bridge the repos with a sport-agnostic **export** (`curate export` in the collector): per-video `start,end` CSVs (the indexer's eval format) + a `manifest.json` that pairs each CSV with its video file. Gate on `is_commercial` so only commercially usable data

ADR-005: Permissive Dependency Stack for the Hosted Monetization Path

Date: 2026-05-30

Status: Accepted

Context

The project will be monetized as a **closed-source hosted API** (users upload videos to a cloud GPU VM and get highlights). Because we never distribute code/binaries, GPL/LGPL distribution copyleft is not triggered ? but **AGPL §13** (network clause) and **metered-API terms** are, and **video-codec patent royalties** are a separate concern from software licenses. A deep-research audit (2026-05-30, 25 adversarially-verified claims against primary sources ? see `docs/MONETIZATION_AUDIT.md`) identified the blockers and clean alternatives. This ADR also **resolves the open licensing question in ADR-001/ADR-004** about WASB.

Decision

1. **Drop `ultralytics`.** Both the library and the `yolov8n.pt` weights are **AGPL-3.0**; §13 makes them a blocker for a closed-source hosted service. Swapping the codebase while keeping the `.pt` does **not** escape ? the weights are AGPL too. Replace player/motion-gating detection with a permissive detector: **torchvision Faster R-CNN / RetinaNet** (BSD, default ? zero new deps, person = COCO class 0), escalating to **YOLOX** or **RT-DETRv2** (Baidu repo, Apache-2.0) if more speed/accuracy is needed. **Ultralytics Enterprise License** is a paid fallback only (quote-only pricing).
2. **Shuttle tracking stays permissive.** **WASB-SBDT** and **TrackNetV3** are MIT (CLEAR for

commercial use)** ? this resolves the licensing unknown flagged in ADR-001/ADR-004. **MonoTrack (Adobe Research, noncommercial) and YOLO-NAS weights (Deci, noncommercial) are banned.** Train our own checkpoints (or confirm per-checkpoint terms), since distributed research weights/datasets aren't covered by the MIT *code* grants.

3. **LLM step ? keep all three strategies open:** (A) eliminate via an offline temporal segmenter (**ActionFormer, MIT** ? the Phase-4 direction), (B) self-host **Qwen2.5-VL-7B/32B (Apache-2.0)** or **InternVL2.5 (MIT)** , or (C) keep **Gemini 2.5 Flash on the paid tier** (required ? the free tier trains on and human-reviews uploaded video). The `google-genai` SDK itself is Apache-2.0 (clear).

4. **Emit royalty-free output.** Standardize highlight output on **AV1 (+ Opus)** to avoid AVC/HEVC encode-side patent exposure; provision an **Ada-class GPU (L4/L40)** for hardware AV1 (NVENC), falling back to **SVT-AV1 (CPU, BSD-3-Clause-Clear)** on T4/A10/A100.

Consequences

- Eliminates the only two hard legal blockers; the rest of the stack (torch, opencv, fastapi, numpy, lapx, ffmpeg-as-subprocess, google-genai SDK) is already permissive.
- Detector swap is contained to `backend/indexer/yolo\_hybrid.py` (the `from ultralytics import YOLO` load + the `model.track(...)` call) plus `config.json` knobs and `tests/test\_yolo\_hybrid.py` mocks. Stage-1 feature outputs (velocity/active-frame/track-id stats) and Stages 2?4 are detector-agnostic and unchanged.
- We now own the frame loop + tracking (ultralytics did both internally). Use a permissive ByteTrack (e.g. `supervision`, MIT, or the MIT ByteTrack repo) on top of `lapx` (already a BSD dep).
- Residual items needing legal sign-off: exact codec-pool royalty thresholds (AVC/HEVC decode of uploads), Ultralytics Enterprise pricing, and per-checkpoint weight licenses ? tracked in `docs/MONETIZATION\_AUDIT.md` §7.
- The `yolo\_hybrid` registry key is now a slight misnomer (no YOLO); keep it for back-compat or rename in a follow-up.

ADR-006: Golden-Set Generation via Vendor Bake-off + a Pluggable Annotation Provider

Date: 2026-05-30

12. user (2026-06-10T05:43:51.662Z)

M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) · *greenlight item*

Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index schema (the conversational seam)

? same row, wired to a store. Fields: `{video\_id(MD5), rally\_ordinal, sport, start, end, ending\_reason/fault\_tags, shot\_count, label\_type, source(human|pseudo|vendor), consent/training\_opt\_in, split(BY MATCH), trajectory\_ref, labeled\_window, derived\_clip\_ref, confidence}`. 3 transcoders (heuristic/LogReg · TAL ActivityNet-JSON · VLM clip?QA), one scorer spine (`metrics.match\_intervals`). Authored in `sports-data-collector` (system-of-record; extends PR #13).

Reserve fault/event taxonomy now. Acceptance: round-trip test; split-by-match enforced; **no Gemini-sourced rows in any training view**.

M0.2 ? Grow the corpus · **DONE (n=6), running** · *owner action + my scoring loop*

6 distinct matches labeled (3 singles incl. Mahadevpura, 1 doubles, 2 multi-court); per-match + LOVO recorded; the

"contrast" data-quality metric validated. Owner adding 1 more varied multi-court-club match ? margin. **Bottleneck =

data + calibration, not method.** Keep growing per new sport.

M0.3 ? Compliance cleanup (**hard blocker ? do before any training run**) · *greenlight item*

- ** (i) PURGE the Gemini-derived TRAINING labels** in `training.label\_candidates` (over Gemini CSVs) ? a **live no-paid-LLM-training violation** that bakes in the moment a training run precedes it. Retarget to human + open-teacher

```

(own-model/WASB) labels; Gemini = eval/reference/fallback only.
- ** (ii) WASB C3 ? first-class action item (owner):** the academic-data checkpoint provenance is
a **commercial-ship
  blocker** that a clean head does NOT launder. Resolve via **own-trained weights** or a clean
detector swap; **multiplies
  per sport.** Carry as a tracked blocker that **gates SHIP** (not Gen-0 research).

```

M0.4 ? Gen-0 TAL harness (FIRST-CLASS ACTION ITEM** per owner) · *greenlight item***

```

Productionize `rally_seq_proto.py` ? a **one-command train?eval?ship-gate** harness over
**ActionFormer (MIT, 1:1
rally=instance)** and/or **ASFormer (MIT, TAS)** (use `sj-li/MS-TCN2` if MS-TCN++, *not* the
Commons-Clause repo),
reusing `metrics.match_intervals`. **Define the ship-gate target up front** (open item): a FLOOR
of "beat the honest n=6
heuristic LOVO **0.480**" is necessary but not sufficient ? set an explicit **F1@tIoU** target
for "very high P/R" (e.g.
F1@tIoU=0.5 for highlights, tighter for precise rally cuts) before declaring victory. Trains on
the GPU/WSL box.

```

====

2.1 Segmenters (under pipeline/segmenters) ?

```

- `@backend/pipeline/segmenters/base.py`
  - `::BasePlaySegmenter` ABC ? `__init__(db, config)`; abstract
  `name`, `description`, `process_video`. $ `process_video(video_id, video_path, player_pool=None,
max_frames=None) -> List[dict{id, start_time, end_time, duration, metadata}]`.
  - `::SegmenterRegistry`, `::segmenter_registry` (singleton). ? **register a new segmenter**:
subclass `BasePlaySegmenter`, call `segmenter_registry.register(MyCls)` at module bottom, AND
import the module in `@backend/pipeline/segmenters/__init__.py` (registration is an IMPORT side-
effect). ? `register()` builds `cls(None, {})` just to read `name` ->
name/description/`__init__` MUST NOT touch `self.db`/`self.config`. ? `list_available()` re-
instantiates + swallows exceptions into "No description available." (a throwing `description` is
masked).
- `@backend/pipeline/segmenters/__init__.py` ? imports all 5
(`motion, gemini, yolo_hybrid, tracknet_hybrid, wasb_hybrid`) = the de-facto registration manifest;
re-exports them + `segmenter_registry` via `__all__`.
- `@backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter` (name `wasb_hybrid`) ?
WSL WASB substrate (MIT; ships pretrained badminton weights, the LIVE default). Imports
`WasbRunner`/`WasbConfig` from `pipeline/detectors/wasb_runner`. Config
`indexing.wasb.velocity_threshold(0.02)` + shared
`dead_time_merge_gap/min_action_window_duration/ai_handoff_padding/ai_max_workers`. ? whole-
video single pass; reuses YOLO-named keys `log_yolo_candidates/store_yolo_candidates`; needs
`{PROVIDER}_API_KEY` (missing -> []); `detection_params` has NO `queue_length` (unlike
tracknet). ? divergent shot_count: `int(shot_count)` inside try with no `is None` short-circuit
(copy-paste drift).
- `@backend/pipeline/segmenters/tracknet_hybrid.py::TrackNetHybridSegmenter` (name
`tracknet_hybrid`) ? copy-paste twin of wasb_hybrid; imports `TrackNetRunner`/`TrackNetConfig`
from `pipeline/detectors/tracknet_runner`; `indexing.tracknet.velocity_threshold(0.02)`;
`detection_params` adds `queue_length`. ? any P3/P4 fix must be applied to BOTH (and arguably
yolo_hybrid).
- `@backend/pipeline/segmenters/yolo_hybrid.py::HybridYoloSegmenter` (name `yolo_hybrid`, no
YOLO remains ? torchvision detector + supervision ByteTrack; ADR-005). `::PERSON_CLASS_ID=1` ?
(ultralytics used 0; wrong value -> zero detections). `::lazy_load_model` (lazy `torchvision`
import so module/tests load without it), `::extract_action_windows_from_chunk`,
`::make_tracker` (per-chunk ByteTrack ? track IDs chunk-local),
`::DETECTOR_FACTORIES`/`::DEFAULT_DETECTOR='fasterrcnn_resnet50_fpn'`. ? velocity at
`effective_fps = fps/frame_skip` -> `yolo_velocity_threshold` coupled to `frame_skip`;
pipelined-vs-sequential P2 (prefetch ffmpeg overlaps sequential GPU);
`yolo_tracker`/`bytetrack.yaml` is dead/back-compat. ? `gemini_*` config keys are deprecated
aliases still honored.
- `@backend/pipeline/segmenters/gemini.py::GeminiPlaySegmenter` (name `gemini`) ? pure-AI, no
P2. ? chunks only if `duration>chunk_duration`; `max_workers=5` hardcoded;
`max_segment_duration=90.0` hardcoded; `debug_duration` silently truncates; `parse_time` accepts
colon timestamps (warns); does NOT persist candidate segments (only hybrids feed the fine-tuning

```

```
table).
- `@backend/pipeline/segmenters/motion.py::MotionPlaySegmenter` (name `motion`) ? pixel
absdiff baseline. ? **metadata + serve/smash/foul/winner events are RANDOM**
(random.sample/randint/uniform/choice); only segmenter using `db.add_event` and passing
server/receiver; uses `print()`; honors `max_frames`.
```

13. user (2026-06-10T05:43:53.704Z)

```
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md:17:- ? **Per-checkpoint
weight/dataset licenses** ? if we borrow pretrained weights (YOLOX, ActionFormer, WASB,
TrackNetV3), the *code* license ? the distributed *weights/dataset* terms. Either train our own
or confirm each checkpoint. **Revisit:** when adopting any borrowed weights.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\MONETIZATION_AUDIT.md:58:- ??
Their *distributed pretrained checkpoints / training datasets* are not covered by the MIT code
grant ? either confirm per-checkpoint terms or **train our own** weights (our roadmap already
plans this).
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HOW_RALLY_DETECTION_WORKS.md:44:*where
the shuttle is*. So they can't teach WASB's detector (that needs `(x,y)` labels). They
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EVALUATION.md:18:> `(x, y)` coordinate
labels, a different and much heavier annotation. Use rally
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EVALUATION.md:20:> coordinate labels if
evaluation shows *detection* (not segmentation) is the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PLATFORM_ARCHITECTURE.md:267:> to train
our owned model would breach their "no competing model" terms (a terms/copyright
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md:71:
(TrackNetV4/WASB ? see ADR-001), which needs coordinate labels.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md:100:-
Only invest in frame-level coordinate labels (for the detector) **if** evaluation shows
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md:175:
itself (no xy labels), only a rally layer on its output.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md:244:2.
**Shuttle tracking stays permissive.** **WASB-SBDT and TrackNetV3 are MIT (CLEAR for commercial
use)** ? this resolves the licensing unknown flagged in ADR-001/ADR-004. **MonoTrack (Adobe
Research, noncommercial) and YOLO-NAS weights (Deci, noncommercial) are banned.** Train our own
checkpoints (or confirm per-checkpoint terms), since distributed research weights/datasets
aren't covered by the MIT *code* grants.
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:125: model trained on provenance-
unclear features **does not launder the provenance.** ROADMAP Phase-4 "train own WASB
```

14. user (2026-06-10T05:44:07.431Z)

```
42:| C3 | **Shuttle-tracking weights provenance** | **P1** | ?? | WASB-SBDT **code is MIT
(clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code
grant. MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) **or**
confirm per-checkpoint terms ?? before commercial deploy. |
```

====

You annotate, per video, the `[start, end)` time range of each **rally**. That's all the evaluation harness needs.

```
> **Important:** these rally timestamps measure and tune **rally segmentation**
> ? i.e. *where* the pipeline says rallies begin and end. They do **not** train
> the shuttle detector (TrackNet/WASB), which learns from per-frame shuttle
> `(x, y)` coordinate labels, a different and much heavier annotation. Use rally
> timestamps as your **golden evaluation set**; only invest in frame-level
> coordinate labels if evaluation shows *detection* (not segmentation) is the
> bottleneck. See `docs/TRACKNET_WSL_SETUP.md`.
```

2. Annotation file format (CSV)

====

Consequences

- The collector becomes the *producer* of labels; the indexer the *consumer* (processing +

- scoring + training). Neither takes a hard dependency on the other beyond the manifest.
- Scoring is free and deterministic (reads the DB; no API). See `docs/EVALUATION.md`.
- Only invest in frame-level coordinate labels (for the detector) **if** evaluation shows **detection**, not **segmentation**, is the bottleneck.

15. user (2026-06-10T05:44:08.406Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md:42:| C3 |
****Shuttle-tracking weights provenance**** | ****P1**** | ? ?? | WASB-SBDT ****code is MIT (clear)****;
distributed ****checkpoints are academic-data-trained**** ? not covered by the MIT code grant.
MonoTrack/YOLO-NAS banned (noncommercial). | Train-own weights (Phase-4 roadmap) ****or**** confirm
per-checkpoint terms ?? before commercial deploy. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md:85: cost, no
external dep) and matches the Phase-4 roadmap (substrate = WASB
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md:220:- `ROADMAP.md`
? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\MONETIZATION_AUDIT.md:64:|
**** (A) Eliminate**** | Offline temporal action localization ? ****ActionFormer**** (MIT), peers MS-TCN
/ ASFormer (verify each) | ****MIT**** | Best long-term margin (no per-call cost, no external
dependency); matches the Phase-4 offline-segmenter roadmap. Most engineering. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\TRACKNET_WSL_SETUP.md:85:4. ****Phase 4 ?**
DB:**** confirmed segments written and linked back to their candidate.**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\ROADMAP.md:61:### Phase 4 ? Learn an
offline, API-free segmenter (NEXT)
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\ROADMAP.md:81:***learned timbre**
classifier ? which, like Phase 4, is ****gated on golden labels****.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\decisions\DECISIONS.md:245:3.
****LLM step ? keep all three strategies open:**** (A) eliminate via an offline temporal segmenter
(****ActionFormer**, MIT ? the Phase-4 direction), (B) self-host ****Qwen2.5-VL-7B/32B**
(Apache-2.0)**** or **InternVL2.5 (MIT)****, or (C) keep ****Gemini 2.5 Flash on the paid tier****
(required ? the free tier trains on and human-reviews uploaded video). The `google-genai` SDK
itself is Apache-2.0 (clear).
C:\Users\avidu\Projects\badminton-highlight-
indexer\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:125: model trained on provenance-
unclear features ****does not launder the provenance.**** ROADMAP Phase-4 "train own WASB

16. assistant (2026-06-10T05:44:09.085Z)

You've hit your session limit · resets 3:50pm (Asia/Calcutta)